Inherited-F4 remap round: run report `run-1`

Status: measured, passed its preregistered rule against its own baseline, superseded by `main` before merge, not merged. Started 2026-10-07T00:30Z (UTC). Study directory: `research/inherited_f4_remap_20261007/`.

What ran

- **Protocol:** `PROTOCOL.md`, written before any confirmatory run. **Runner:** `run.sh` (thin shell around `ecbench exec`, Valgrind Callgrind, `taskset`, `jq`).
- **Baseline:** `ecbench` from `main` at `a3652eddafb949073bf0f96b210a75b671cfb346`, SHA-256 `7e6ad0d2…`. A byte-identical copy served as the A/A arm.
- **Candidate:** the baseline plus `results/run-1/candidate.diff` (SHA-256 `3779a2b0…`), SHA-256 `7bfdd8c9…`. It was built from a working tree and was never committed, because a Conductor reservation required by `CLAUDE.md` could not be taken. A lint-only revision is kept as `results/run-1/final.diff`.
- **Host:** Intel Xeon @ 2.80 GHz cloud VM, 4 logical CPUs, 16 GiB, Linux 6.18 (kernel build `fc-v77`), BMI2 present; `rustc` 1.97.0; Valgrind 3.22.0 (`results/run-1/host.txt`).
- **Inputs:** 60 exact `ecbench` child inputs exported from frozen specs (`results/run-1/inputs.sha256`):
  - ladder: m = 13 (16), m = 19 (16), m = 23 (4);
  - round-1 holdouts: m = 13 (8), m = 19 (4);
  - fresh holdouts: m = 13 (8, target seed 202610071013), m = 19 (4, target seed 202610071019).
- **Arms:** the inherited-F4 and F6-IC oracles of the F6-IC ladder, unchanged.

Runs requested and completed

measurement	requested	completed	note
identical-output gate	60 inputs × 2 binaries	60 × 2	all 60 `IDENTICAL` (`gate.tsv`)
Callgrind total Ir	44 inputs × 2	44 × 2	`ir.tsv`, `ir.raw.tsv`
pinned CPU time, ABAB with A/A	3 inputs × 5 rounds × 4	55 of 60 rows	worker restart killed the run in L23 seq 25 round 4 (`wall.tsv`)
L23 seq 25 re-run, new kernel build `fc-v80`	5 rounds × 4	20 of 20	`wall-L23-rerun.tsv`, same binaries

Measured: instructions (Callgrind Ir, whole process)

One run per binary and input; Callgrind counts are deterministic. The ratio is candidate / baseline.

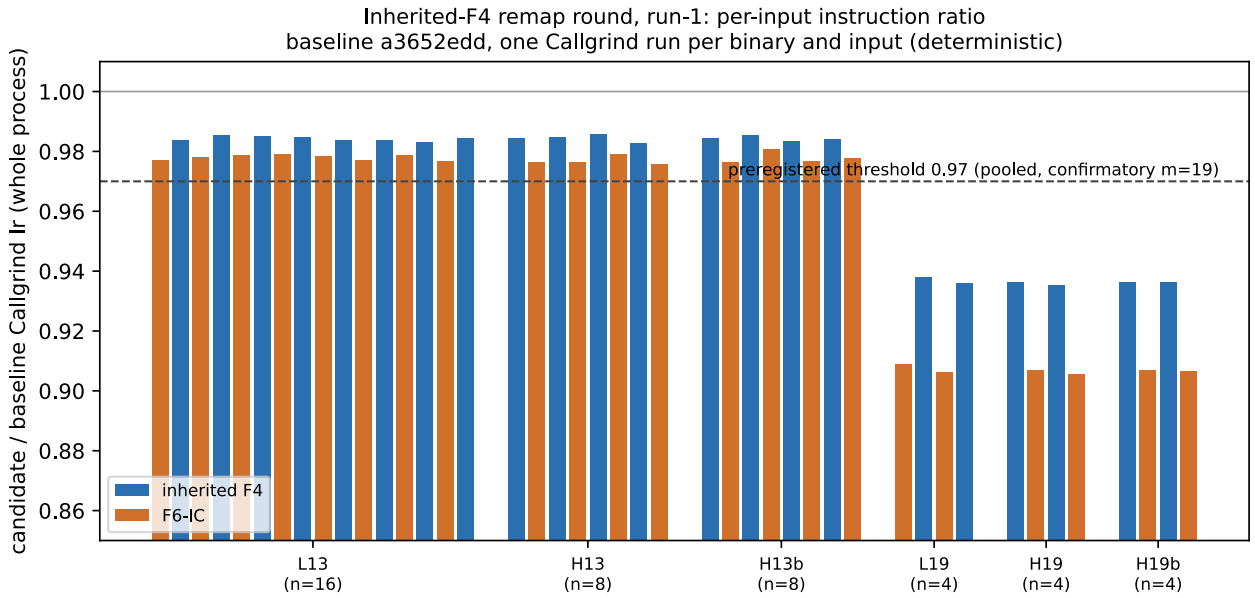


Figure 1: Per-input ratio from `results/run-1/ir.tsv`, run-1, 44 inputs. Blue: inherited F4; orange: F6-IC. Dashed: the preregistered threshold on the pooled confirmatory m = 19 ratio.

m, arm	inputs	pooled ratio (derived: sum cand / sum base)	per-input range

13, inherited F4	16	0.984	0.983–0.986
13, F6-IC	16	0.978	0.976–0.981
19, inherited F4	6	0.936	0.935–0.938
19, F6-IC	6	0.907	0.905–0.909

Decision statistic (derived): pooled over the confirmatory m = 19 inputs (L19 seq 24, 27, 28, H19, H19b), 423,612,106,635 / 458,788,722,509 = 0.9233. The preregistered threshold was ≤ 0.97. The largest per-input ratio is 0.986 (H13-4).

Measured: pinned process CPU time (practicality note)

`taskset -c 3`, run alone on the host. Each round runs baseline, candidate, the A/A copy, then the candidate again. Scheduling delay ≤ 7.3 ms per run.

input	baseline median (s)	candidate median (s)	A/B median (range); A/A median (range)
L19 seq 25, inherited F4	7.13	6.44	0.908 (0.889–0.919); 0.994 (0.982–1.008)
L19 seq 24, F6-IC	4.68	4.05	0.866 (0.838–0.877); 1.006 (0.976–1.030)
L23 seq 25, rounds 1–4, kernel `fc-v77`	90.0	78.1	0.866 (0.860–0.870); 1.000 (0.995–1.008)
L23 seq 25, re-run, kernel `fc-v80`	91.7	78.8	0.860 (0.841–0.873); 0.993 (0.979–1.016)

Superseding check (2026-10-11)

`ecbench` built from `main` at `7666b708e` (SHA-256 `6ae1a2a1…`), which contains 1646 (a word-wise `pext`/`pdep` rewrite for every map) and 1671 (eager row rewrites). It was measured on three of the same inputs (`results/main-7666b708-check.tsv`). Its counted outputs differ from the round-2 baseline (`solver_ops` higher on all three; F6-IC gate statistics differ on L19 seq 24), so this is not a gated A/B.

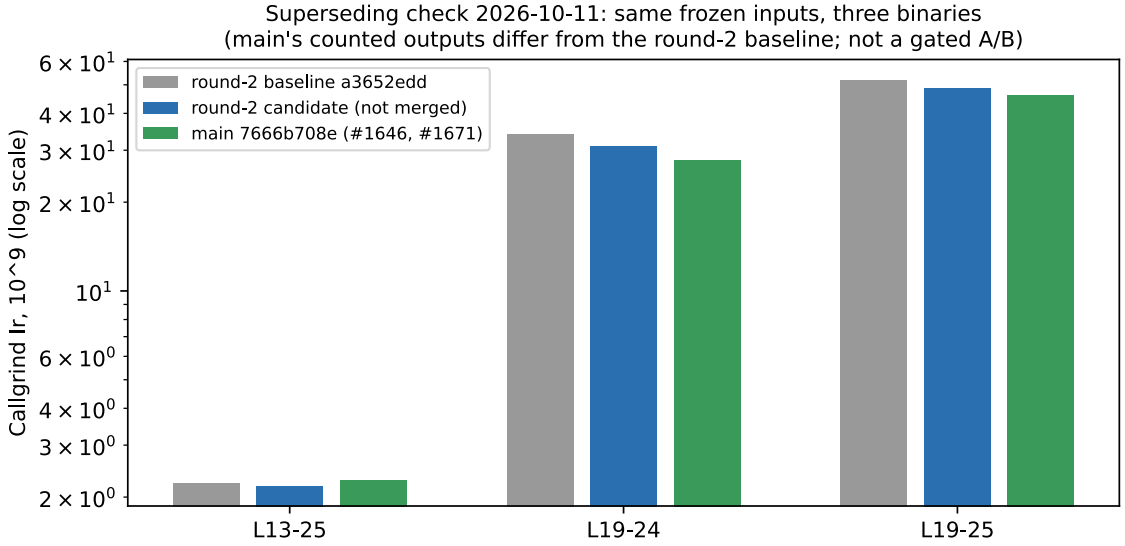


Figure 2: Raw Callgrind Ir of the round-2 baseline, the round-2 candidate and `main` `7666b708e` on L13 seq 25, L19 seq 24 and L19 seq 25.

input	round-2 candidate Ir	main Ir	main / candidate (derived)
L13 seq 25	2,185,880,355	2,286,390,878	1.046
L19 seq 24	30,866,373,759	27,799,877,737	0.901

L19 seq 25	48,778,989,164	46,233,505,868	0.948

Raw records

`results/run-1/`: `gate.tsv`, `ir.tsv`, `ir.raw.tsv`, `ir/*.err`, `wall.tsv`, `wall-L23-rerun.tsv`, `inputs/`, `host.txt`, `binaries.sha256`, `candidate.diff`, `final.diff`, `lint-followup.tsv`; `results/main-7666b708-check.tsv`. The graph source is `plot.py` in this directory, which reads only those files.